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PROJECT SYNOPSIS

Automated Attribution of Unknown Cryptocurrency Wallets to Nearest Virtual Asset Service Providers (VASPs) through Blockchain Intelligence APIs

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1. Executive Summary

The rapid proliferation of Virtual Digital Assets (VDAs) and decentralized blockchain ecosystems has created significant new challenges for law enforcement investigations globally. When a suspect cryptocurrency wallet address is identified during a cyber fraud, ransomware, or money laundering investigation, law enforcement agencies (LEAs) must determine which Virtual Asset Service Provider (VASP) — such as a centralized exchange or custodial wallet provider — has custody of or control over those funds. Without this attribution, it is impossible to freeze assets, identify beneficial owners, or compel disclosure under lawful authority.

This project proposes the development of an **Automated Blockchain Intelligence & VASP Attribution Engine** — a full-stack, production-grade software platform that automates the process of tracing blockchain transaction flows from any unknown suspect wallet address and identifies the nearest centralized exchange, custodial wallet provider, or VASP receiving direct deposits from that wallet. The system integrates directly with investigative portals through a secure REST API and is capable of generating court-admissible investigation reports within minutes of receiving a wallet address.

The platform supports six major blockchain networks — Bitcoin (BTC), Ethereum (ETH), Tron (TRX), BNB Chain (BSC), Solana (SOL), and Polygon (MATIC) — and employs a layered intelligence approach that combines blockchain graph analytics, heuristic clustering algorithms, a supervised machine learning ensemble, and a large language model (LLM) powered generative analysis layer for explainable risk assessment. Risk scoring, typology detection, and automated VASP attribution are performed entirely without human intervention, delivering actionable intelligence to investigators in real time.

2. Problem Background & Context

2.1 The Rise of Cryptocurrency-Enabled Cybercrime

The global adoption of cryptocurrencies and decentralized finance (DeFi) platforms has created a parallel financial system frequently exploited by cybercriminals. Key crime categories affecting India and global jurisdictions include:

- **Ransomware Attacks:** Threat actors demand cryptocurrency ransom payments, routing proceeds through mixers and cross-chain bridges before depositing at exchanges.
- **Investment Scams & Pig Butchering:** Fraudsters collect victim funds into aggregator wallets before laundering proceeds.
- **Darknet Marketplace Proceeds:** Vendors convert illicit earnings through layered intermediary wallets.
- **Terrorist Financing:** Small, structured cryptocurrency payments are used to fund activities while evading AML controls.
- **Cyber Fraud Proceeds:** Phishing, SIM-swap fraud, and business email compromise funds are converted to cryptocurrency for cross-border laundering.

2.2 The Unhosted Wallet Attribution Problem

A critical bottleneck in cryptocurrency investigations is the prevalence of **unhosted wallets** — self-custodied wallet addresses not maintained by any registered VASP. When a suspect wallet is identified, investigators face three scenarios:

1. **Direct exchange wallet:** Easily identifiable via public address tagging databases.
2. **Intermediate/relay wallet:** A wallet that receives and forwards funds — attribution requires multi-hop tracing to find the receiving exchange.
3. **Mixer/tumbler/bridge output:** Funds are deliberately obfuscated through privacy-enhancing services before reaching an exchange.

The current manual process for scenarios 2 and 3 requires specialized blockchain analysts, proprietary intelligence platforms (e.g., Chainalysis, Elliptic), and several days of analysis. This delay significantly reduces the probability of successful asset freezing, as cryptocurrency transfers are irreversible and funds can be withdrawn from exchanges within hours of deposit.

2.3 Jurisdictional & Regulatory Context

Under the Prevention of Money Laundering Act (PMLA), IT Act 2000, and the evolving Virtual Digital Asset (VDA) regulatory framework in India, LEAs are empowered to compel VASPs registered with the Financial Intelligence Unit-India (FIU-IND) to freeze wallets and disclose beneficial owner information. However, this authority can only be exercised when the responsible VASP is correctly identified — a prerequisite that the current investigative infrastructure cannot reliably deliver at scale.

3. Existing Challenges in Investigation Workflows

3.1 Manual Blockchain Tracing

Current investigation workflows require human analysts to manually:

- Query block explorer websites (Etherscan, BscScan, TronScan) for transaction histories.
- Follow transaction chains hop-by-hop through multiple intermediate wallets.
- Cross-reference addresses against proprietary intelligence databases.
- Determine VASP attribution based on pattern recognition and institutional knowledge.

This process is time-intensive (typically 2–10 days per wallet), error-prone, and not scalable to the volume of cryptocurrency-related cases filed annually.

3.2 Incomplete Intelligence Coverage

Existing public intelligence databases (e.g., Bitcoin Abuse, community-sourced exchange wallet lists) cover only a fraction of the known exchange ecosystem and are rarely updated in real time. Attribution of deposit wallet addresses — temporary addresses assigned by exchanges for individual user deposits — is particularly difficult as these addresses do not appear in public intelligence databases.

3.3 Multi-Chain Complexity

Modern money laundering operations exploit cross-chain bridges (e.g., Stargate, Wormhole, LayerZero) to move funds between blockchains, defeating single-chain tracing tools. An investigation that begins on Ethereum may need to follow funds through BNB Chain before reaching a TRON-based exchange deposit address.

3.4 Volume & Scalability Constraints

Indian LEAs collectively file thousands of cryptocurrency-related complaints annually. Existing tools require manual invocation per case and cannot process investigations at scale. An automated, queue-based system capable of handling hundreds of simultaneous trace jobs is required to meet operational demands.

4. Proposed Solution Overview

The **Blockchain Intelligence & VASP Attribution Engine** is a distributed software platform designed to automate every stage of the cryptocurrency attribution workflow:

4.1 Core Capabilities

Capability	Description
Automated Wallet Tracing	BFS-based multi-hop transaction graph construction from any suspect wallet
Multi-Chain Support	Bitcoin, Ethereum, Tron, BNB Chain, Solana, Polygon
VASP Attribution	Shortest-path nearest-VASP identification with confidence scoring
Address Clustering	Common-Input Ownership (Bitcoin) and deposit-pattern (EVM/TRX/SOL) heuristics
Risk Scoring	Weighted multi-factor risk score (0–100) with band classification
Typology Detection	Layering, peel chains, smurfing, fan-out, fan-in, mixer interaction
ML Wallet Classification	Supervised ensemble model for wallet suspicion scoring
LLM Generative Analysis	Gemini 2.5 Flash powered natural language fraud analysis
PDF Report Generation	Court-ready investigation reports with audit trail
Portal Integration	REST API integration with law enforcement investigative portals
Real-Time Alerts	WebSocket-based high-risk wallet alerts with Redis pub/sub
Case Management	Multi-user RBAC-protected investigation case tracking

4.2 High-Level Workflow

1. Suspect wallet address is submitted through the portal dashboard.
2. Multi-Chain Adapter Layer fetches transactions (Etherscan / TronScan / SolScan / BscScan / PolygonScan / Mempool).
3. BFS Transaction Graph Builder constructs a NetworkX DiGraph up to 10 hops / 10,000 nodes.
4. Intelligence Tagging Layer checks all nodes against Known VASP DB + Deposit Sweep Detector + ML Fallback.
5. Nearest VASP Finder computes shortest-path from seed to each tagged VASP node and ranks by hop count.
6. Risk Scoring + Typology Detection applied on the full graph.
7. LLM Analysis Layer (Gemini 2.5 Flash) generates natural language analysis.
8. PDF Report automatically generated with SHA-256 audit hash.
9. Investigator routes freeze/disclosure request via the portal dashboard.

5. System Architecture

5.1 Technology Stack

Layer	Technology	Purpose
Backend API	FastAPI (Python 3.11)	Async REST API server

Layer	Technology	Purpose
Task Queue	Celery 5.4 + Redis	Async background trace jobs
Database	PostgreSQL (SQLAlchemy 2.0 async)	Primary data store
Cache	Redis	Blockchain API response caching (TTL: 30 min)
Graph Engine	NetworkX 3.3	Transaction graph construction & analysis
ML Framework	scikit-learn 1.3	Wallet suspicion classification model
LLM Integration	Google Gemini 2.5 Flash	Generative fraud analysis & NLP risk narratives
Secondary LLM	Mistral AI	Alternative generative analysis engine
PDF Rendering	WeasyPrint + Jinja2	Investigation report PDF generation
Object Storage	AWS S3 / MinIO	Report PDF artifact storage
Frontend	Next.js 14 (TypeScript)	Investigator dashboard
Graph Visualization	vis-network	Interactive transaction graph visualization
Authentication	JWT (HS256) + bcrypt	Role-based access control

5.2 Service Decomposition

The backend is organized into clearly separated domain modules:

```
app/
■■■ adapters/ — Blockchain API clients (per-chain adapters)
■■■ api/ — FastAPI routers (REST endpoints)
■■■ auth/ — JWT authentication & RBAC
■■■ audit/ — Tamper-evident audit log
■■■ cases/ — Investigation case management
■■■ clustering/ — Address clustering heuristics & attribution
■■■ graph/ — Transaction graph builder & VASP finder
■■■ intel/ — Intelligence database lookup & tagging
■■■ ml/ — Wallet suspicion ML model
■■■ reports/ — PDF report generation
■■■ risk/ — Risk scoring, alerts, LLM analysis
■■■ sahyog/ — Portal integration client
■■■ tasks/ — Celery async task definitions
■■■ typology/ — Money laundering pattern detectors
■■■ vasp_db/ — VASP address database
```

5.3 Asynchronous Processing Model

1. **POST /cases/{case_id}/traces** → Creates a TraceJob row with status=queued, dispatches run_trace Celery task.
2. Celery worker executes BFS graph construction (up to 5-minute deadline, 10,000-node limit).
3. Progress updates are written to the TraceJob row (0–100%) and broadcast via WebSocket.
4. On completion, VASP matching, risk scoring, typology detection, and report generation are executed.
5. High-risk results (score ≥ 70) trigger immediate RiskAlert creation and Redis pub/sub broadcast.

6. Multi-Chain Blockchain Data Collection Layer

6.1 Adapter Architecture

The system implements a pluggable BlockchainAdapter abstract base class that defines a uniform interface for all blockchain data sources. Each concrete adapter handles chain-specific API peculiarities while exposing a consistent normalized data model.

Adapter	Chain	API Source	Supported Data
EtherscanAdapter	ETH	Etherscan V2 API	Native ETH + ERC-20 token transfers (merged)
BscScanAdapter	BSC	BscScan API	BNB + BEP-20 token transfers
PolygonScanAdapter	MATIC	PolygonScan API	MATIC + ERC-20 transfers
TronScanAdapter	TRX	TronScan API	TRX + TRC-20 token transfers
SolScanAdapter	SOL	Solscan Pro API	SOL + SPL token transfers
MempoolAdapter	BTC	Mempool.space API	Native Bitcoin UTXO transactions

6.2 Normalized Data Model

Every adapter produces RawTransaction objects with a uniform schema regardless of chain. Key fields include: tx_hash, from_addr, to_addr, amount (Decimal), fee, timestamp, chain identifier, block_height, is_bridge flag, and bridge_protocol. This normalization enables the graph analytics engine to operate identically across all supported chains.

6.3 Resilience & Caching

- **Exponential backoff retry** (up to 2 attempts, base delay 0.5s, max 5s) for transient HTTP errors and rate limit responses (HTTP 429/503).
- **Redis transaction cache** with configurable TTL (default 30 minutes) — cache key: blockchain:{chain}:{address}:{page}:{page_size}.
- **Raw API response archival** — every successful upstream API response is SHA-256 hashed and written to disk for audit trail purposes, without blocking the trace.
- **Health check probes** — each adapter can perform a lightweight upstream connectivity test using well-known probe addresses.

6.4 ERC-20 & Token Transfer Support

The Ethereum, BSC, and Polygon adapters fetch both native currency transactions and token transfers separately, then merge and deduplicate by transaction hash. Token transfer records take precedence over native transaction records for the same hash, ensuring the graph reflects token-denominated value flows (stablecoins, USDT, USDC) rather than the often-zero ETH value of token transfer transactions.

7. Transaction Graph Analytics Engine

7.1 BFS-Based Graph Construction

The core of the attribution engine is a **Breadth-First Search (BFS) transaction graph builder** that constructs a directed NetworkX DiGraph from a seed suspect wallet address.

Parameter	Default	Description
max_hops	5	Maximum BFS depth (hard limit: 10)
min_nodes	20	Minimum target wallet nodes before BFS terminates
deadline_seconds	300 (5 min)	Wall-clock timeout
max_per_address	200	Top-N transactions per address (by amount, preventing fan-out explosion)
max_frontier_size	150	BFS frontier cap (top-N addresses by cumulative volume)
NODE_COUNT_LIMIT	10,000	Hard cap on graph size

BFS process:

1. Seed address is added to the graph and frontier queue.
2. At each hop, the adapter fetches transactions for every frontier address.
3. Edges are added to the DiGraph with transaction attributes (amount, fee, timestamp, chain, bridge flag).
4. Addresses flagged as VASPs by the vasp_check_fn callback are tagged and excluded from further expansion (branch-stop optimization).
5. When EDGE_COUNT_PRUNE_THRESHOLD (50,000 edges) is exceeded, low-value edges below the 10th percentile by amount are pruned.

7.2 Graph Expansion

Beyond the initial trace, investigators can request an expand_graph operation that deepens an existing graph by one additional BFS hop from all leaf nodes (addresses with no outgoing edges within the current graph). This enables iterative investigation without restarting from scratch.

7.3 Graph Persistence

Completed NetworkX DiGraph objects are serialized to JSON using networkx.node_link_data() and persisted to the trace_graphs database table alongside the TraceJob record. Graph JSON is also stored in S3/MinIO object storage for long-term retention and retrieval.

7.4 Live Feed Monitoring

During active BFS traces, the system provides a real-time address feed endpoint (GET /traces/{trace_id}/live-feed) that tails the Celery worker log to extract the most recently fetched blockchain addresses. This powers a live investigation loading screen in the frontend, giving investigators real-time visibility into trace progress.

8. Address Clustering & Attribution Pipeline

8.1 Common-Input Ownership (CIO) Heuristic — Bitcoin

For Bitcoin transaction graphs, the platform applies the industry-standard **Common-Input Ownership (CIO) heuristic**: any two addresses that co-sign inputs to the same transaction are inferred to be controlled by the same entity.

1. For each node n in the graph, collect predecessors = [addr for addr in G.predecessors(n)].
2. If len(predecessors) ≥ 2, add undirected edges between every predecessor pair in a co-spend auxiliary graph.

3. Compute connected components of the co-spend graph using NetworkX.
4. Each component with ≥ 2 addresses is emitted as a Cluster.

8.2 Deposit Address Pattern Heuristic — EVM/TRX/SOL

For EVM chains, Tron, and Solana, the platform uses a **hub-and-spoke deposit address pattern heuristic**: exchange deposit wallets exhibit a characteristic behavior where they receive funds from a user and immediately forward 95–100% of those funds to a known exchange hot wallet.

Sweep detection criteria:

- Node has exactly 1 outgoing edge to a known exchange hot wallet.
- The sweep ratio (outflow / inflow) ≥ 0.95.
- No other significant outgoing flows.

Detected deposit addresses are tagged with the parent exchange name and a confidence score of 0.85–0.95.

> **Example:** An unlabelled address receives 1.0 ETH and within minutes forwards 0.998 ETH to a Binance hot wallet. The system identifies this as a Binance deposit address with ~90% confidence and tags it accordingly in the graph.

8.3 Attribution Confidence Scoring

For each cluster-to-VASP attribution, the system computes a weighted confidence score:

```
Confidence = (0.4 x address_match_ratio
+ 0.3 x volume_similarity
+ 0.2 x behavioral_similarity
+ 0.1 x temporal_proximity) x 100
```

- **address_match_ratio:** Fraction of cluster addresses found in the VASP DB.
- **volume_similarity:** Cosine similarity of volume profiles between cluster and known VASP activity.
- **behavioral_similarity:** Pattern-match score against known VASP behavioral fingerprints.
- **temporal_proximity:** Recency weight (recent activity scores higher).

8.4 Typology Tagging

Attributions to mixer, bridge, or darknet VASPs automatically emit TypologyTag database records, triggering escalated risk handling and inclusion in generated investigation reports.

9. Machine Learning & AI Intelligence Layer

9.1 Nearest VASP Finder Algorithm

The core VASP attribution algorithm (find_nearest_vasps) operates through a three-stage pipeline:

Stage 1 — Intelligence Tagging: All graph nodes are checked against the known_addresses intelligence database. Tags include entity name, entity type (CEX, DEX, mixer, bridge, darknet, sanctioned, scam, ransomware), intelligence source, and confidence score.

Stage 2 — Deposit Sweep Detection + ML Fallback: Unlabeled nodes are checked for deposit sweep patterns. Remaining unlabeled nodes undergo ML-based probabilistic classification: feature vectors are extracted from the graph, the trained ensemble model predicts entity type probabilities, and nodes with CEX probability > 0.70 are tagged as probabilistic VASP matches with XAI reasoning attached.

Stage 3 — Shortest Path Ranking: For each tagged VASP node, networkx.shortest_path() finds the minimum-hop path from the seed address. Results are ranked by: hops (ascending) → total value transferred (descending) → confidence (descending). A hop penalty (max(0, (hops-1) × 0.10)) is applied to the confidence score to reflect uncertainty introduced by intermediate wallets.

9.2 Wallet Suspicion Classification Model

The platform includes a trained **wallet suspicion ensemble model** for probabilistic classification of unindexed wallet addresses.

Architecture:

- RobustScaler for outlier-robust feature normalization.
- RandomForestClassifier (100 estimators, max depth 10, balanced class weights).
- HistGradientBoostingClassifier (100 iterations, max depth 6, L2 regularization 0.3).
- VotingClassifier — soft voting ensemble of RF + HGB.
- CalibratedClassifierCV (isotonic calibration, 5-fold CV) for realistic posterior probabilities.

Feature Set (25 domain-specific features):

Category	Features
Volume	total_tx_count, in_tx_count, out_tx_count, in_out_tx_ratio, total_inflow, total_outflow, net_balance_retention
Statistical	mean_tx_amount, std_tx_amount, max_tx_amount, amount_gini, round_amount_ratio, dust_tx_ratio
Temporal	lifespan_hours, tx_per_day, burst_volume_ratio, rapid_forwarding_ratio, night_activity_ratio
Topological	in_degree, out_degree, counterparty_dispersion, fan_in_ratio, fan_out_ratio, reciprocity_ratio
Exposure	high_risk_entity_exposure

Key behavioral indicators:

- **Rapid Forwarding Ratio:** % of funds forwarded within 60 minutes of arrival — high values indicate transit/relay wallet behavior consistent with money laundering layering.
- **Net Balance Retention:** Ratio of remaining balance to total inflow — near-zero retention is characteristic of exchange deposit addresses and peeling chains.
- **Burst Volume Ratio:** Fraction of total volume concentrated in short time bursts — structuring and smurfing typologies produce characteristic burst patterns.
- **Amount Gini Coefficient:** Statistical inequality measure of transaction amounts — mixers and tumblers produce distinctive Gini profiles.

9.3 Explainable AI (XAI) Risk Reasoning

For each wallet suspicion score, the model generates human-readable explanations identifying the most contributing features and detected behavioral patterns. Each explanation includes the feature name, measured value, severity level (high/medium/low), and a plain-language description. Detected pattern labels (e.g., Rapid Pass-Through / Peeling Transit, Zero Balance Retention) are included in generated investigation reports to assist non-technical investigators and satisfy judicial scrutiny requirements.

9.4 LLM-Powered Generative Analysis

The platform integrates **Google Gemini 2.5 Flash** for generative analysis of transaction graphs. The LLM receives a compact summary of the most significant wallets (up to 60, selected by graph degree centrality) and generates:

- Per-wallet suspicion scores (0–100) with natural language reasoning.
- Classification labels: clean, suspicious, highly_suspicious, unknown.
- Detected behavioral flags (e.g., mixer_interaction, rapid_layering, large_outflow).

A secondary **Mistral AI** integration provides an alternative generative analysis layer for comparative verification and redundancy. Results are cached in-process per trace to avoid redundant API calls. The system degrades gracefully when LLM API keys are not configured, returning empty analysis rather than blocking the trace.

10. Risk Scoring & Typology Detection Engine

10.1 Multi-Factor Risk Scoring Formula

The platform computes an objective, integer risk score in [0, 100] using a weighted multi-factor formula:

```
Risk Score = round(
(0.35 x direct_exposure
+ 0.20 x indirect_exposure
+ 0.20 x vasp_risk_category
+ 0.15 x typology_flags
+ 0.10 x volume_anomaly) x 100
)
```

Factor	Weight	Description
direct_exposure	35%	Ratio of flagged (mixer/sanctioned/darknet/scam) direct neighbors
indirect_exposure	20%	Same ratio computed over all nodes within 3 hops
vasp_risk_category	20%	Risk tier of nearest attributed VASP
typology_flags	15%	Normalized count of active laundering typology flags
volume_anomaly	10%	Z-score normalized transaction velocity (clamped to [0,1])

VASP Risk Tier Lookup:

VASP Type	Risk Weight	Rationale
CEX (regulated exchange)	0.1	Lowest risk — KYC/AML regulated
DEX	0.3	Medium-low — no KYC but onchain
Bridge	0.5	Medium — cross-chain obfuscation risk
Mixer/Tumbler	0.9	Very high — designed for obfuscation
Darknet	1.0	Maximum risk — illicit marketplace

Automatic Override: Any sanctioned or darknet address anywhere in the transaction graph forces the score to ≥ 90, ensuring immediate escalation when OFAC or darknet exposure is confirmed.

10.2 Risk Band Classification

Band	Score Range	Recommended Action
Low	0 – 39	Monitor
Medium	40 – 69	Initiate disclosure request
High	70 – 100	Immediate freeze request

10.3 Money Laundering Typology Detectors

The platform implements a suite of pattern detectors for known money laundering typologies:

- **Layering Detector:** Identifies long simple paths (≥ 5 hops) with no address reuse — characteristic of the second stage of money laundering. Confidence: min(0.6 + (path_len - 5) × 0.05, 1.0).
- **Peel Chain Detector:** Identifies linear sequences of hops where each address retains a small 'dust' amount and forwards the remainder. Detection requires minimum path length ≥ 4, monotonically decreasing amounts with minimum 80% forwarding ratio.
- **Smurfing / Structuring Detector:** Identifies addresses receiving large-value inflows and splitting them into many small-value outflows (fan-out ratio ≥ 3× fan-in, each outflow < 40% of average inflow).
- **Fan-Out Detector:** Identifies hub addresses distributing funds to many counterparties simultaneously — characteristic of exchange hot wallets or aggregator wallets.
- **Fan-In Detector:** Identifies consolidation addresses receiving funds from many independent sources — characteristic of exchange deposit address aggregation.
- **Mixer Interaction Detector:** Identifies direct or indirect interactions with known mixer or tumbler addresses in the intelligence database.

Each detector returns a DetectionResult with a match_confidence score and a subgraph containing pattern-matched nodes for inclusion in investigation reports.

10.4 Real-Time Risk Alerts

When a trace produces a risk score in the high band (≥ 70), the system:

1. Creates a RiskAlert database record with the trace ID, case ID, wallet address, score, and timestamp.
2. Publishes a JSON broadcast to the risk_alerts_broadcast Redis channel.
3. Frontend clients subscribed via WebSocket receive immediate notification.

Significant risk score changes (> 10 points between assessments) are separately published to the risk_score_changes Redis channel and written to the audit log.

11. Report Generation & Investigative Intelligence

11.1 Investigation-Ready PDF Reports

The platform automatically generates comprehensive PDF investigation reports using WeasyPrint (CSS-based PDF renderer) and Jinja2 HTML templates. Reports are designed to meet the documentation standards required for court submissions and regulatory disclosures.

Report sections:

1. **Case Metadata:** Case ID, title, description, investigation officer, organizational unit, creation date.
2. **Wallet Summary:** All suspect wallet addresses with balance, transaction count, first/last activity, chain.

3. **VASP Attribution:** Ranked list of identified VASPs with: VASP name, entity type, confidence score, hop distance, path from suspect wallet, total value transferred, intelligence source.

4. **Risk Assessment:** Composite risk score, band classification, contributing factor breakdown, recommended action (monitor / disclose / freeze).

5. **Typology Findings:** Detected laundering patterns with confidence scores and pattern descriptions.

6. **Transaction Timeline:** Chronological event log of significant transactions.

7. **Audit Trail:** Report SHA-256 hash, generation timestamp, generated-by user, supervisor signature field.

8. **Low-Confidence Attributions:** Probabilistic ML-based VASP matches with XAI reasoning (clearly labeled as probabilistic rather than confirmed).

11.2 Report Integrity & Chain of Custody

Each generated report is:

- SHA-256 hashed and the hash stored alongside the report record.
- Written to S3/MinIO object storage with server-side encryption.
- Linked to an immutable audit log entry recording the generating user and timestamp.
- Pending supervisor signature before status transitions to 'signed'.

11.3 Supervisor Approval Workflow

A sign_report API endpoint allows supervisors to countersign reports with their digital identity, transitioning report status from pending_signature to signed. Both the signing event and the supervisor identity are captured in the tamper-evident audit log, creating a complete chain of custody for evidentiary use.

11.4 Recommended Action Routing

The report includes a machine-generated recommended_action field based on risk score:

- Score < 40 → monitor (continue surveillance)
- Score 40–69 → disclose (file disclosure request to VASP)
- Score ≥ 70 → freeze (immediate asset freezing request)

This recommendation, combined with the identified VASP attribution, allows investigators to directly route the appropriate lawful disclosure or freezing request to the correct entity.

12. Investigation Portal & Dashboard

12.1 Frontend Architecture

The investigator-facing portal is built with **Next.js 14** (TypeScript) using the App Router pattern. The frontend communicates exclusively with the backend REST API using JWT Bearer token authentication.

View	Functionality
Landing Page	Authentication, system overview
Investigation Workspace	Primary trace submission, real-time progress, graph visualization
Fund Flow Visualization	Interactive vis-network directed graph with node coloring by entity type
Trace Report Page	Full investigation report viewer with VASP attribution, risk scores, typology findings

View	Functionality
Dashboard	Aggregate statistics, recent cases, risk alert feed
Case Management	Case creation, wallet assignment, status tracking

12.2 Real-Time Graph Visualization

The fund flow visualization component renders the complete NetworkX DiGraph in the browser using vis-network, with:

- **Node coloring** by entity type: red (mixer/darknet), orange (bridge/DEX), green (CEX), grey (unknown).
- **Edge sizing** proportional to transaction value.
- **Interactive exploration:** click nodes to view transaction details, expand subgraphs.
- **Path highlighting:** the minimum-hop path from suspect wallet to nearest VASP is highlighted.

12.3 Case Management & RBAC

The platform implements a three-tier role-based access control (RBAC) model:

Role	Permissions
Investigator	Create/view cases, submit traces, view reports
Supervisor	All investigator permissions + sign reports + view all cases in their unit
Admin	Full system access + user management + intelligence DB admin

Cases are scoped to organizational units, ensuring investigative data is isolated between agencies. Supervisors within an org unit can view all cases in their unit; investigators see only their own cases by default.

12.4 Portal Integration (SAHYOG)

The platform includes a SAHYOGClient and SAHYOGService that implement a structured API integration with the investigative portal ecosystem:

- **Payload construction:** Case metadata, attributed VASP details, report content hash, wallet addresses.
- **Authentication:** Supports both API key and OAuth2 bearer token auth modes (configurable per deployment).
- **Submission tracking:** SAHYOGSubmission ORM records track submission status, portal reference numbers, and timestamps.
- **Automated routing:** On report sign-off, the system can automatically construct and submit a lawful disclosure request payload to the correct VASP via the portal.

13. Security, Audit & Compliance Framework

13.1 Authentication & Authorization

- **JWT tokens** (HS256) with 8-hour access token expiry and 30-day refresh tokens.
- **Key rotation:** JWT signing keys rotate every 90 days.
- **bcrypt password hashing** with configurable cost factor.
- **Column-level database encryption** for sensitive wallet address and case data fields.
- All API endpoints require at minimum require_investigator dependency injection.

13.2 Rate Limiting

The API implements per-IP rate limiting (default: 100 requests/minute) with grace handling that returns HTTP 429 with Retry-After headers rather than silently dropping requests.

13.3 Tamper-Evident Audit Log

Every significant platform action generates an AuditLogEntry record capturing:

- Actor user ID and role.
- Action type (case_created, trace_submitted, report_generated, report_signed, SAHYOG_submitted, etc.).
- Resource ID (case, trace, report UUID).
- Before/after state snapshots (JSON).
- Timestamp (UTC).

Audit log entries are append-only and reference-integrity protected — no application code path deletes or modifies audit records. Maximum audit log size is configurable (default: 10 GB).

13.4 Security Headers

The SecurityHeadersMiddleware enforces:

- Strict-Transport-Security (HSTS)
- X-Content-Type-Options: nosniff
- X-Frame-Options: DENY
- Content-Security-Policy
- X-XSS-Protection

13.5 Raw API Response Archival

Every upstream blockchain API response is SHA-256 hashed and archived to disk. This archive provides:

- Proof that transaction data used in attribution was retrieved from a genuine upstream API (not fabricated).
- Reproducibility — the trace can be re-executed against the same archived data.
- Evidence trail for court admissibility challenges.

14. Scalable Infrastructure & Deployment Architecture

14.1 Deployment Architecture

The platform is containerizable and deployable across multiple cloud and on-premise configurations. The core deployment topology:

- **Load Balancer / Nginx** — reverse proxy and TLS termination in front of all services.
- **FastAPI (Uvicorn / Gunicorn workers)** — handles all REST API requests.
- **Next.js Frontend** — static/SSR investigator portal.
- **Celery Worker Pool** — multiple workers for parallel trace execution.
- **PostgreSQL** — primary database (streaming replication for production).
- **Redis** — Celery broker + API response cache.
- **S3 / MinIO** — report PDFs and raw API response archives.

14.2 Cloud Deployment Configuration

The platform ships with a render.yaml deployment manifest configured with:

- Web service: FastAPI + Celery worker (combined process via start.sh).
- PostgreSQL: External managed database (e.g., Supabase).
- Redis: External managed Redis (e.g., Upstash).
- S3: External object storage (AWS S3 or self-hosted MinIO).
- Alembic migrations run as pre-deploy command.

For production law enforcement deployments, the platform supports fully air-gapped on-premise deployment with internal PostgreSQL cluster (streaming replication), on-premise Redis Sentinel, MinIO object storage cluster, and internal Nginx reverse proxy with TLS termination.

14.3 Horizontal Scalability

- Additional Celery worker instances can be added without code changes.
- TRACE_QUEUE_DEPTH_LIMIT (default: 100) prevents queue saturation.
- Each worker handles one run_trace task at a time with configurable soft time limit (450s) and hard kill limit (540s).
- Graph size limits (NODE_COUNT_LIMIT = 10,000, EDGE_COUNT_PRUNE_THRESHOLD = 50,000) enforce bounded resource consumption per trace.

14.4 Database Schema

The platform employs PostgreSQL with Alembic-managed schema migrations. Core tables:

Table	Purpose
users	Investigator accounts with RBAC roles
org_units	Organizational unit isolation
cases	Investigation cases with status tracking
case_wallets	Wallet addresses assigned to cases
trace_jobs	Async trace job tracking with progress
trace_graphs	Serialized NetworkX graph JSON
vasp_matches	VASP attribution results per trace
attributions	Cluster-to-VASP attribution records
clusters	Address cluster records
typology_tags	Detected laundering pattern tags
risk_alerts	High-risk wallet alert records
reports	Generated investigation report records
sahyog_submissions	Portal submission tracking
audit_log	Tamper-evident action audit trail
known_addresses	VASP intelligence address database

14.5 Performance Characteristics

Metric	Target
Trace job throughput	100 concurrent trace jobs (configurable queue depth)
BFS graph construction	≤ 5 minutes per trace (300s deadline)
API response time	< 200ms for read endpoints (cached)
Report generation	< 30 seconds per PDF report
Blockchain API cache hit rate	> 70% (30-minute TTL, common addresses recur across traces)
Node count ceiling	10,000 nodes per trace graph

15. Expected Outcomes & Impact

15.1 Investigative Efficiency Gains

Metric	Current State	With Platform
Time to VASP attribution	2–10 business days	< 10 minutes (automated)
Blockchain chains supported	1–2 (manual per tool)	6 (simultaneous)
Concurrent investigations	5–10 (analyst capacity)	100+ (queue-based)
Report generation	Hours (manual drafting)	< 30 seconds (automated)
Error rate in attribution	High (manual analysis)	Low (algorithmic + ML)

15.2 Asset Freezing Effectiveness

The primary operational benefit of the platform is the dramatic reduction in time-to-freeze. Cryptocurrency exchange withdrawal processing typically takes minutes to hours. Current manual attribution workflows often exceed exchange withdrawal windows, allowing suspects to liquidate assets before freeze orders can be filed. With sub-10-minute automated attribution, LEAs gain the capability to file timely asset freezing requests in the majority of investigations.

15.3 Coverage Expansion

By combining multiple intelligence layers, the platform achieves significantly higher attribution coverage than any single method alone:

- **Known-address database:** ~60–70% coverage of major exchange hot wallets.
- **Deposit sweep detection:** Adds ~15–20% coverage for exchange deposit addresses.
- **ML probabilistic classification:** Adds ~5–10% coverage for novel/unlisted wallets with high behavioral similarity to known VASPs.
- **Combined coverage:** Estimated 80–90% of suspect wallets traceable to a VASP within 3 hops.

15.4 Cross-Border Investigation Support

The multi-chain architecture directly supports cross-border cybercrime investigations where proceeds flow across jurisdictions and blockchains. By maintaining a unified graph view across BTC, ETH, TRX, BSC, SOL, and MATIC within a single investigation case, analysts can track funds through complex cross-chain laundering schemes without switching tools or losing chain-of-custody.

15.5 Strengthening Regulatory Compliance

Automated, standardized attribution reports with machine-verifiable content hashes strengthen the evidentiary value of blockchain evidence in court proceedings. The structured report format, supervisor approval workflow, and immutable audit trail align with requirements for legally admissible digital evidence under Indian Evidence Act standards.

15.6 Future Roadmap

The platform architecture supports planned extensions including:

- **Cross-chain bridge tracing:** Automatic detection and traversal of cross-chain bridge transactions (e.g., following USDT from Ethereum → Tron via bridge).
- **DeFi protocol classification:** Identification of Uniswap, Curve, Aave, and other DeFi protocol interactions within transaction graphs.
- **Real-time wallet monitoring:** Configurable alerts for new transactions from suspect wallets during active investigations.
- **Network-wide analytics:** Aggregate intelligence across all investigations to identify emerging laundering typologies and new VASP clusters.
- **Mobile investigator app:** Lightweight mobile interface for field investigators to submit wallet addresses and receive quick attribution results.
- **Federated intelligence sharing:** Secure, privacy-preserving sharing of VASP intelligence between LEA organizations.

16. Conclusion

The Automated Blockchain Intelligence & VASP Attribution Engine represents a comprehensive, production-ready solution to the attribution gap that currently impedes cryptocurrency-related law enforcement investigations. By automating multi-hop transaction graph tracing, VASP attribution, risk scoring, typology detection, and report generation into a unified platform accessible to non-specialist investigators, the system directly addresses the operational needs of LEAs handling the rapidly growing volume of cryptocurrency-related cybercrime cases.

The platform's layered intelligence approach — combining deterministic heuristics, supervised machine learning, and LLM-powered generative analysis — provides both high coverage and high confidence in attribution results, while the explainable AI components ensure that algorithmic conclusions can withstand judicial scrutiny. The secure, auditable architecture satisfies evidentiary chain-of-custody requirements, and the scalable Celery-based task queue enables the system to grow with investigation volumes without architectural changes.

By reducing VASP attribution time from days to minutes, the platform directly improves asset freezing success rates, strengthens beneficial owner identification, and enhances the effectiveness of cross-border cryptocurrency crime investigations involving Virtual Digital Assets.